

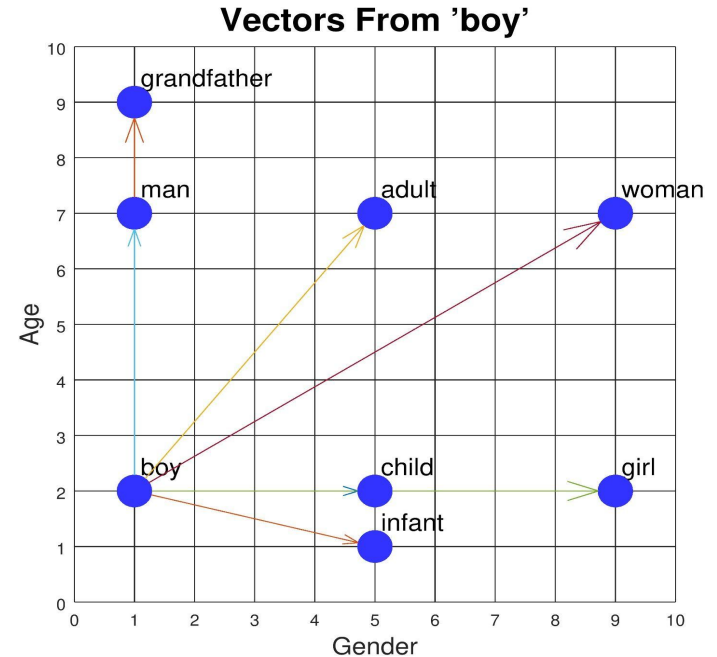


RAG and Information Retrieval

Solving LLMs limitations by providing facts for grounded answers

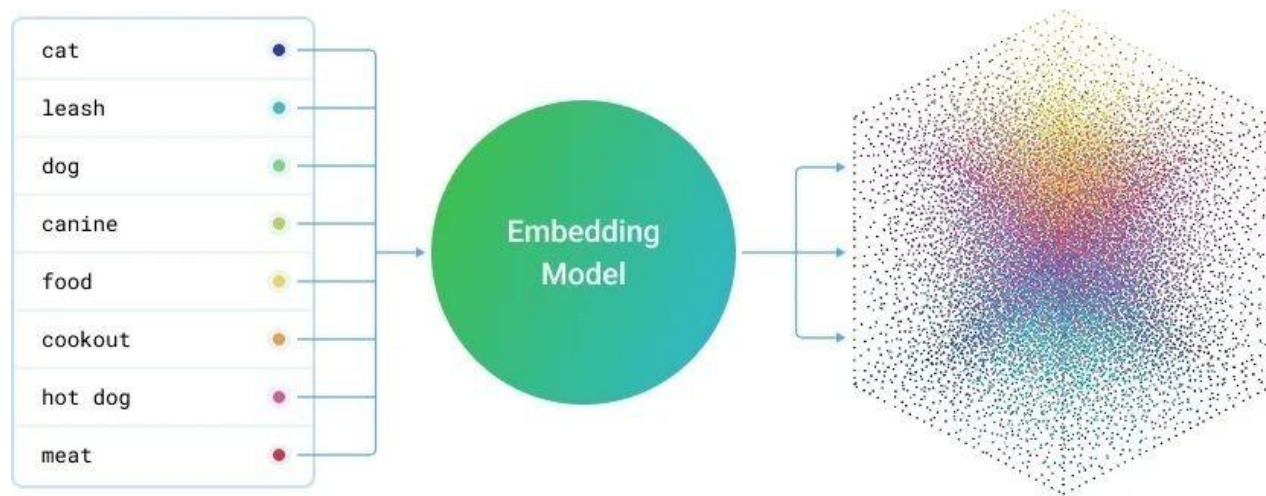
Embeddings

An embedding is a vector representation of something, where similar things should have similar vectors



The Problem: Text Is Not Numeric

- Machine learning models need numbers, not raw words.
- The key question: how do we convert words, sentences, or documents into useful numbers?





Bag of Words

- A simple representation: count how many times each vocabulary word appears.

vocabulary	dog	cat	car	food
sentence	2	0	0	1

"dog dog food" becomes [2, 0, 0, 1]

- The vector is easy to interpret.
- Each dimension corresponds to one word in the vocabulary

Bag of Words

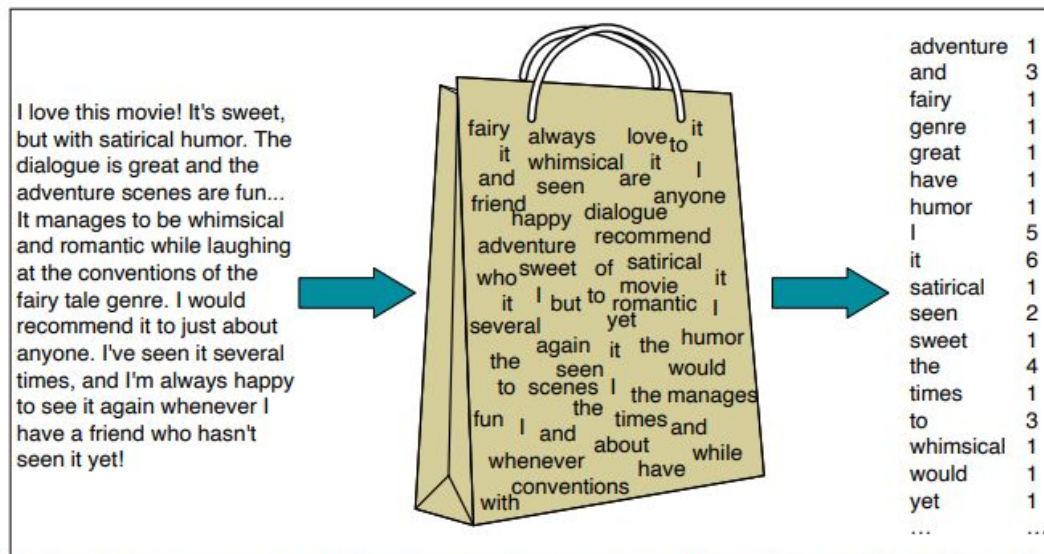
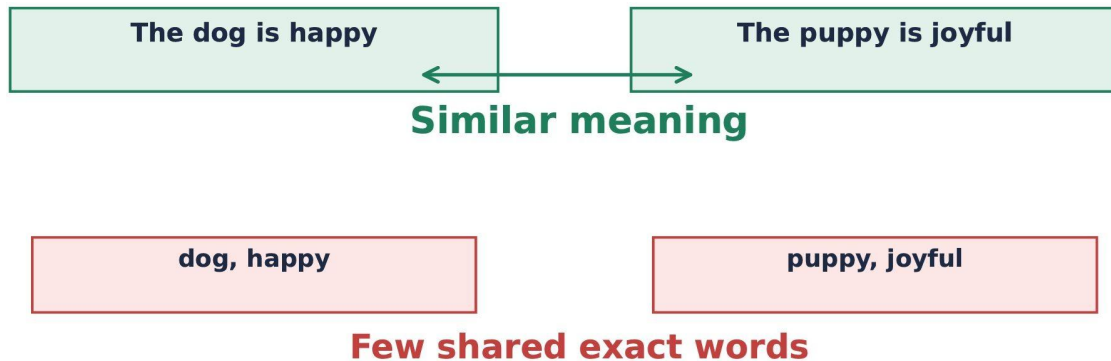


Figure 11.2 Intuition of the classic vector space model applied to a single document. The position of the words is ignored (the *bag-of-words* assumption) and we make use of the frequency of each word.



The Limitation

- Counting exact words does not mean understanding meaning.



TF-IDF

- Still counts words, but gives more weight to informative words.
- Intuition: a word is important if it appears often in this document but not in every document.



$$\text{tf}_{t,d} = \begin{cases} 1 + \log_{10} \text{count}(t,d) & \text{if } \text{count}(t,d) > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\text{idf}_t = \log_{10} \frac{N}{\text{df}_t}$$

$$\text{tf-idf}(t,d) = \text{tf}_{t,d} \cdot \text{idf}_t$$

Cons apart, we can make vectors with this

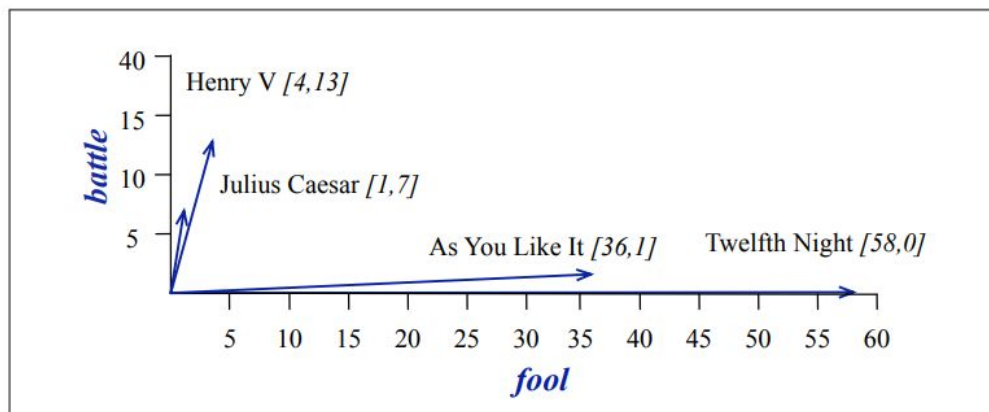
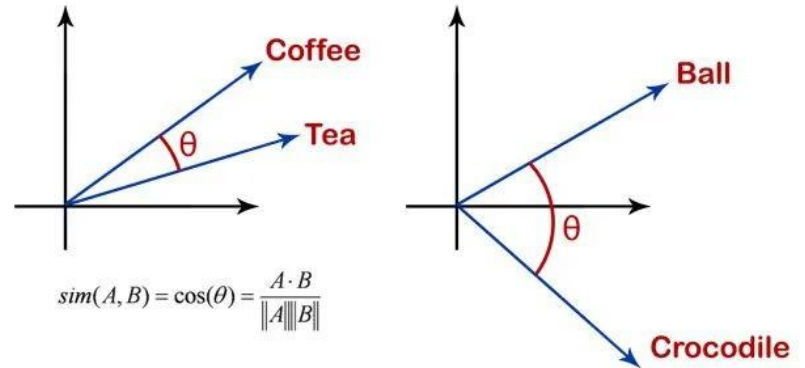


Figure 11.5 A spatial visualization of the document vectors for the four Shakespeare play documents, showing just two of the dimensions, corresponding to the words *battle* and *fool*. The comedies have high values for the *fool* dimension and low values for the *battle* dimension.

“Two documents that are similar will tend to have similar words, and if two documents have similar words their column vectors will tend to be similar.” [1, ch. 11, p. 5]

Cosine Similarity

- Once text is a vector, we need a way to compare vectors.
- If two vectors point in a similar direction, cosine similarity is high.





From Manual Vectors to Learned Vectors

**Bag of Words
TF-IDF**



Embeddings

Manual Features:

- Counts
- Visible vocabulary dimensions
- Exact word overlap

Learned Features:

- Dense vectors
- Trained from data
- Can capture useful similarity

Note: This connects to neural networks, hidden layers also learn useful internal representations.



What Embeddings Are Used For

Search

find documents similar to a query

Recommendations

find similar products, songs, movies, or users

Clustering

group similar texts or items

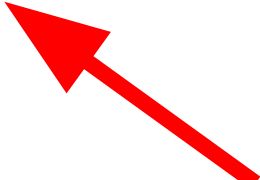
Classification

use embeddings as input features

RAG

retrieve relevant context before answering

They are useful when similarity or meaning matters.



Retrieval Augmented Generation

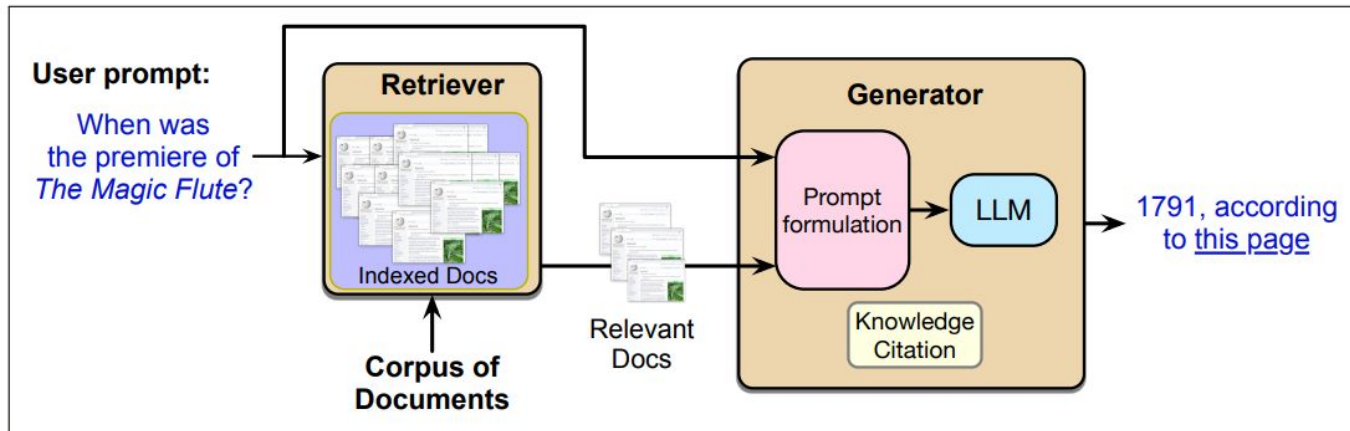


Figure 11.13 Retrieval-augmented generation takes as input a user prompt (which may express an information need like this question example), and a corpus of documents that may be useful in meeting the information need. The method has two stages: **retrieval**, which returns relevant documents from the collection, and **generation**, in which an LLM **generates** text given the documents as a prompt. Some generations include a **knowledge citation** that can help the user decide whether to trust the generation, or follow up if they are interested.



References

D. Jurafsky & J. H. Martin, **Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition, with Language Models** (Chapter 11, 3rd ed. draft, 2026).

Available at: <https://web.stanford.edu/~jurafsky/slp3/>